

Upendra Singh

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Professional Summary

Computer Science and Engineering undergraduate with strong problem-solving skills and hands-on experience in developing scalable software applications. Passionate about building efficient solutions through software engineering, machine learning, and full-stack development, with a strong foundation in core computer science concepts and collaborative development.

Education

Pranveer Singh Institute of Technology, Kanpur B.Tech in Computer Science and Engineering	2023 – 2027 Percentage: 76.65% (through Semester V)
S.V.M Inter College, Orai Class XII, Science PCM	2021 – 2022 Percentage: 78.8%
S.V.M Inter College, Konch Class X	2019 – 2020 Percentage: 82.33%

Technical Skills

Programming Languages: C++, Python, JavaScript, SQL
Frontend: HTML, CSS, React.js, Responsive Design
Backend: Node.js, Express.js, Flask,
Databases: MySQL, MongoDB, Database Design, SQL Queries
Developer Tools: Git, GitHub, Docker, Postman, VS Code
Machine Learning: Generative AI, Natural Language Processing (NLP), Variational Autoencoders (VAE), Data Preprocessing
Core Subjects: Data Structures and Algorithms, Object-Oriented Programming, DBMS, Operating Systems, Computer Networks

Projects

GenTwin: AI-Driven Digital Twin for Cybersecurity Gap Detection Python, VAE, Generative AI, Digital Twin, SWaT Dataset	Feb 2026 – Present
- Developed an AI-driven digital twin system to detect cybersecurity gaps in industrial control systems. - Implemented Variational Autoencoders (VAE) to generate realistic cyberattack scenarios using the SWaT dataset. - Generated and analyzed 10,000+ simulated industrial control system records to identify abnormal system behavior. - Applied machine learning and data preprocessing techniques to improve anomaly detection and vulnerability analysis.	
CareSync: Smart Hospital Appointment System Python, Flask, OpenCV, MySQL, Twilio, REST APIs	Nov 2025 – Feb 2026
- Built a smart hospital appointment management system with patient registration, scheduling, and authentication modules. - Developed 8+ REST APIs using Flask for appointment booking, patient data management, and admin operations. - Integrated MySQL database to store patient records, appointment details, and authentication data securely. - Implemented face recognition and OTP-based verification to improve patient identity validation.	
RunX: Multi-Language Online Compiler HTML, CSS, JavaScript, Node.js, Judge0 API	Jan 2025 – May 2025
- Created a web-based online compiler supporting code execution in 10+ programming languages. - Integrated Judge0 API for secure code compilation, execution, and real-time output generation. - Built backend request-handling modules using Node.js to process code submissions and API responses. - Designed a responsive user interface to improve code writing, input handling, and output visualization.	

Certifications

- Complete Web Development Course** – Udemy
- Database Management System and SQL** – Infosys Springboard
- Data Science in Python** – Great Learning

Achievements

- Finalist** – Cybersecurity Hackathon 2025, C3iHub IIT Kanpur
- Solved 250+ Data Structures and Algorithms problems** on LeetCode
- 5-Star Gold Badge** in C++ and Problem Solving on HackerRank